EXP 801 DS Wheat Heat Drought

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ANATOMICAL DATA - IMPORT

- Importing anatomical data
- Some data is collected on the individual stomata level, while other traits are aggregated to a pot level

Changes in v2: removing outliers

Load the libraries

```
library(knitr)
#knitr::spin("Exp801.anat.import.v2.r")
library(tidyverse)
library(dae)
```

Set up constants and files

```
opts_chunk$set("tidy" = FALSE, "dev" = "png", "dpi" = 300, fig.path = "figures/",
               "fig.width" = 7.5, "fig.height" = 10, "comment"=NA, warning = FALSE)
options(width = 90)
```

Import indv.dat from WU analysis, as this will have all of the design terms I need

```
load("../../data/indv.v2.Rdata")


importFile <- "../data/Stomatal Anatomy Data.xlsx"
saveFile <- "../data/anat.dat.v2.Rdata"
```

Anatomical traits

- First import the traits that were measured at the pot level
- This sheet contains the treatment and rep info
- Use that sheet to then import the traits measured at the stomata level

```
responses.anat <- c("StomatalCount", "StomatalDensity", "StomatalArea",
                    "StomatalWidth", "StomatalLength", "PoreArea", "gsmax")
respunits.anat <- c("", "per mm^2", "microm^2", "microm","microm", "microm", "mol m^-2 s^-1")
```

```
anat.pot.dat <- readxl::read_xlsx(importFile, "Processed Stomatal anatomy Data") %>% as.data.frame()
names(anat.pot.dat)[c(7, 8, 9, 11, 13, 15, 17)] <- responses.anat
summary(anat.pot.dat)
```

Import pot level data

```
      Filename        Line          Temperature       Watering        Rep       
 Length   :159   Min.   : 1.000   Length   :159   Length   :159   Min.   :1.000  
 N.unique :159   1st Qu.: 3.000   N.unique :  2   N.unique :  2   1st Qu.:1.500  
 N.blank  :  0   Median : 6.000   N.blank  :  0   N.blank  :  0   Median :3.000  
 Min.nchar: 42   Mean   : 5.509   Min.nchar:  4   Min.nchar:  7   Mean   :2.503  
 Max.nchar: 65   3rd Qu.: 8.000   Max.nchar:  7   Max.nchar:  9   3rd Qu.:3.500  
```

```
                Max.   :10.000                                     Max.   :4.000
 Line_Temperature_Watering_Rep StomatalCount   StomatalDensity  StomatalArea
 Length   :159                 Min.   :16.00   Min.   :31.11   Min.   :1045
 N.unique :159                 1st Qu.:23.00   1st Qu.:44.72   1st Qu.:1517
 N.blank  :  0                 Median :27.00   Median :52.50   Median :1624
 Min.nchar: 16                 Mean   :26.35   Mean   :51.22   Mean   :1640
 Max.nchar: 22                 3rd Qu.:29.00   3rd Qu.:56.38   3rd Qu.:1751
                               Max.   :41.00   Max.   :79.72   Max.   :2726
 Stdev Stomatal Area (µm^2) StomatalWidth   Stdev Stomatal Width (µm) StomatalLength
 Min.   :116.2              Min.   :27.94   Min.   :1.697             Min.   :48.11
 1st Qu.:185.2              1st Qu.:31.67   1st Qu.:3.090             1st Qu.:59.95
 Median :235.9              Median :34.09   Median :3.850             Median :62.91
 Mean   :250.8              Mean   :34.20   Mean   :3.977             Mean   :62.36
 3rd Qu.:300.6              3rd Qu.:35.79   3rd Qu.:4.637             3rd Qu.:65.83
 Max.   :882.9              Max.   :50.81   Max.   :9.158             Max.   :72.90
 Stdev Stomatal Length (µm)    PoreArea      Stdev pore area (µm^2)     gsmax
 Min.   : 2.297             Min.   : 542.4   Min.   :  70.17        Min.   :0.8403
 1st Qu.: 4.528             1st Qu.: 844.9   1st Qu.: 127.19        1st Qu.:1.2128
 Median : 5.263             Median : 921.7   Median : 152.02        Median :1.3498
 Mean   : 5.696             Mean   : 913.6   Mean   : 171.93        Mean   :1.3419
 3rd Qu.: 6.401             3rd Qu.:1010.4   3rd Qu.: 188.63        3rd Qu.:1.4710
 Max.   :22.882             Max.   :1240.4   Max.   :1290.05        Max.   :2.0044
```

```
anat.stoma.dat <- readxl::read_xlsx(importFile, "Raw All Output Anatomy Data") %>% as.data.frame()
names(anat.stoma.dat)[c(2:5, 8)] <- c("Stomata", responses.anat[3:6])
summary(anat.stoma.dat)
```

Import stomata level data

```
      Filename       Stomata       StomatalArea  StomatalWidth   StomatalLength
 Length   :4189   Min.   : 1.00   Min.   : 428   Min.   :14.85   Min.   : 31.85
 N.unique : 159   1st Qu.: 7.00   1st Qu.:1411   1st Qu.:30.59   1st Qu.: 57.27
 N.blank  :   0   Median :14.00   Median :1598   Median :33.41   Median : 62.29
 Min.nchar:  42   Mean   :14.06   Mean   :1625   Mean   :34.08   Mean   : 61.95
 Max.nchar:  65   3rd Qu.:20.00   3rd Qu.:1796   3rd Qu.:36.75   3rd Qu.: 66.48
                  Max.   :41.00   Max.   :6075   Max.   :77.95   Max.   :193.08
 Pore length (µm) Pore width (µm)     PoreArea      Pore depth (µm)
 Min.   : 17.29   Min.   : 8.647   Min.   : 234.9   Min.   : 4.324
 1st Qu.: 31.10   1st Qu.:15.548   1st Qu.: 759.4   1st Qu.: 7.774
 Median : 33.82   Median :16.911   Median : 898.5   Median : 8.456
 Mean   : 33.64   Mean   :16.819   Mean   : 902.4   Mean   : 8.409
 3rd Qu.: 36.10   3rd Qu.:18.049   3rd Qu.:1023.5   3rd Qu.: 9.025
 Max.   :104.84   Max.   :52.420   Max.   :8632.6   Max.   :26.210
```

```
##incorporate treatment info into anat.stoma.dat
anat.stoma.dat <- left_join(anat.stoma.dat,
                            anat.pot.dat %>% select(Filename, Line, Temperature,
                                                    Watering, Rep,
                                                    Line_Temperature_Watering_Rep))
```

```
Joining with `by = join_by(Filename)`
```

```
anat.pot.dat$Replicate <- factor(anat.pot.dat$Rep)
```

```
anat.pot.dat$Line.ID <- sprintf("L%02d", anat.pot.dat$Line)
anat.pot.dat$Line.ID <- factor(anat.pot.dat$Line.ID)
anat.pot.dat$Rm_Temperature <- anat.pot.dat$Temperature
anat.pot.dat$Rm_Temperature[anat.pot.dat$Temperature == "Heat"] <- "7,Heat"
anat.pot.dat$Rm_Temperature[anat.pot.dat$Temperature == "Ambient"] <- "8,Ambient"
anat.pot.dat$Rm_Temperature <- factor(anat.pot.dat$Rm_Temperature)
anat.pot.dat$Watering <- factor(anat.pot.dat$Watering)

anat.pot.dat <- left_join(anat.pot.dat %>% select(-Line), indv.dat,
                          by = c("Watering", "Replicate", "Rm_Temperature", "Line.ID"))

##do the same for the anat.stoma.dat dataframe
anat.stoma.dat$Replicate <- factor(anat.stoma.dat$Rep)
anat.stoma.dat$Line.ID <- sprintf("L%02d", anat.stoma.dat$Line)
anat.stoma.dat$Line.ID <- factor(anat.stoma.dat$Line.ID)
anat.stoma.dat$Rm_Temperature <- anat.stoma.dat$Temperature
anat.stoma.dat$Rm_Temperature[anat.stoma.dat$Temperature == "Heat"] <- "7,Heat"
anat.stoma.dat$Rm_Temperature[anat.stoma.dat$Temperature == "Ambient"] <- "8,Ambient"
anat.stoma.dat$Rm_Temperature <- factor(anat.stoma.dat$Rm_Temperature)
anat.stoma.dat$Watering <- factor(anat.stoma.dat$Watering)
anat.stoma.dat$Stomata <- factor(anat.stoma.dat$Stomata)

anat.stoma.dat <- left_join(anat.stoma.dat %>% select(-Line), indv.dat,
                          by = c("Watering", "Replicate", "Rm_Temperature", "Line.ID"))
```

Define variables to allow for appending design info from indv.dat

Remove outliers

Removing outliers detected through analysis

```
anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER7A09" &
               anat.stoma.dat$Stomata == 30,
               c("StomatalArea", "StomatalLength",
                   "StomatalWidth", "PoreArea")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER7A12" &
                 anat.stoma.dat$Stomata == 19,
               c("StomatalArea", "StomatalWidth")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER7C04" &
                 anat.stoma.dat$Stomata == 24,
               c("StomatalArea", "StomatalLength",
                 "PoreArea")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER7C07" &
                 anat.stoma.dat$Stomata == 39,
               c("StomatalArea", "StomatalLength",
                 "PoreArea")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER7C11" &
                 anat.stoma.dat$Stomata == 38,
               c("StomatalArea", "StomatalLength",
                 "PoreArea")] <- NA
```

```
anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER7D02" &
                 anat.stoma.dat$Stomata == 29,
               c("StomatalArea", "StomatalLength",
                 "StomatalWidth", "PoreArea")] <- NA


anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER7E14" &
                 anat.stoma.dat$Stomata == 21,
               c("StomatalArea", "StomatalLength",
                 "PoreArea")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER8A03" &
                 anat.stoma.dat$Stomata == 32,
               c("StomatalArea","StomatalWidth")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER8C02" &
                 anat.stoma.dat$Stomata == 30,
               c("StomatalArea", "StomatalLength",
                 "PoreArea")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER8C05" &
                 anat.stoma.dat$Stomata == 28,
               c("StomatalArea")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER8C11" &
                 anat.stoma.dat$Stomata == 20,
               c("StomatalArea", "StomatalLength",
                 "PoreArea")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER8E02" &
                 anat.stoma.dat$Stomata == 15,
               c("StomatalArea", "StomatalLength",
                 "PoreArea")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER8E05" &
                 anat.stoma.dat$Stomata == 19,
               c("StomatalArea")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER8E12" &
                 anat.stoma.dat$Stomata == 27,
               c("StomatalArea", "StomatalWidth")] <- NA

anat.stoma.dat[anat.stoma.dat$Cell.ID == "CER8F10" &
                 anat.stoma.dat$Stomata == 28,
               c("StomatalArea")] <- NA
```

Finalise data

```
anat.pot.dat <- anat.pot.dat[order(anat.pot.dat$Cell.ID),]
summary(anat.pot.dat)
```

```
      Filename      Temperature       Watering       Rep
 Length   :159   Length   :159   Limited  :79   Min.   :1.000
```

```
N.unique :159   N.unique :  2   Unlimited:80   1st Qu.:1.500
N.blank  :  0   N.blank  :  0                  Median :3.000
Min.nchar: 42   Min.nchar:  4                  Mean   :2.503
Max.nchar: 65   Max.nchar:  7                  3rd Qu.:3.500
                                               Max.   :4.000

Line_Temperature_Watering_Rep StomatalCount   StomatalDensity  StomatalArea
Length   :159                 Min.   :16.00   Min.   :31.11   Min.   :1045
N.unique :159                 1st Qu.:23.00   1st Qu.:44.72   1st Qu.:1517
N.blank  :  0                 Median :27.00   Median :52.50   Median :1624
Min.nchar: 16                 Mean   :26.35   Mean   :51.22   Mean   :1640
Max.nchar: 22                 3rd Qu.:29.00   3rd Qu.:56.38   3rd Qu.:1751
                              Max.   :41.00   Max.   :79.72   Max.   :2726

Stdev Stomatal Area (µm^2) StomatalWidth   Stdev Stomatal Width (µm) StomatalLength
Min.   :116.2              Min.   :27.94   Min.   :1.697             Min.   :48.11
1st Qu.:185.2              1st Qu.:31.67   1st Qu.:3.090             1st Qu.:59.95
Median :235.9              Median :34.09   Median :3.850             Median :62.91
Mean   :250.8              Mean   :34.20   Mean   :3.977             Mean   :62.36
3rd Qu.:300.6              3rd Qu.:35.79   3rd Qu.:4.637             3rd Qu.:65.83
Max.   :882.9              Max.   :50.81   Max.   :9.158             Max.   :72.90

Stdev Stomatal Length (µm)    PoreArea      Stdev pore area (µm^2)     gsmax
Min.   : 2.297             Min.   : 542.4   Min.   :  70.17        Min.   :0.8403
1st Qu.: 4.528             1st Qu.: 844.9   1st Qu.: 127.19        1st Qu.:1.2128
Median : 5.263             Median : 921.7   Median : 152.02        Median :1.3498
Mean   : 5.696             Mean   : 913.6   Mean   : 171.93        Mean   :1.3419
3rd Qu.: 6.401             3rd Qu.:1010.4   3rd Qu.: 188.63        3rd Qu.:1.4710
Max.   :22.882             Max.   :1240.4   Max.   :1290.05        Max.   :2.0044

Replicate    Line.ID     Rm_Temperature      Cell.ID          Pot.ID      Room    Lane
1:40      L01    :16   7,Heat   :79     Length   :159   Length   :159   CER7:79   A:28
2:39      L02    :16   8,Ambient:80     N.unique :159   N.unique :159   CER8:80   B:24
3:40      L03    :16                    N.blank  :  0   N.blank  :  0             C:27
4:40      L05    :16                    Min.nchar:  7   Min.nchar:  6             D:28
          L06    :16                    Max.nchar:  7   Max.nchar:  6             E:24
          L07    :16                                                             F:28
          (Other):63
   Position      PCoord    Side    Location  Block  SubUnit      Cell        Treatment
2      :12   1      :  8   N:79   Inner:80   1:80   1:80    1      :16   Length   :159
3      :12   2      :  8   S:80   Outer:79   2:79   2:79    2      :16   N.unique : 40
4      :12   3      :  8                                   3      :16   N.blank  :  0
5      :12   4      :  8                                   4      :16   Min.nchar:  3
6      :12   5      :  8                                   5      :16   Max.nchar:  3
9      :12   6      :  8                                   6      :16
(Other):87   (Other):111                                  (Other):63
                        Line        AWatering      HWatering
SUNCHASER                 :16   rest     :79   rest     :80
ROCKSTAR                  :16   Limited  :40   Limited  :39
PBI17N006-0N-0N-89N       :16   Unlimited:40   Unlimited:40
PBIC15034-0C-24N-010N-1N-0N:16
PBI16C009-0C-0N-0N-010N-14N:16
PBI19N007-0N-3N           :16
(Other)                   :63
```

```
                          ALine                                HLine                      TW
 rest                        :79   rest                        :80   Ambient-Limited  :40
 SUNCHASER                   : 8   SUNCHASER                   : 8   Ambient-Unlimited:40
 ROCKSTAR                    : 8   ROCKSTAR                    : 8   Heat-Limited     :39
 PBI17N006-0N-0N-89N         : 8   PBI17N006-0N-0N-89N         : 8   Heat-Unlimited   :40
 PBI17N013-0C-0N-0N-010N-8N  : 8   PBIC15034-0C-24N-010N-1N-0N: 8
 PBIC15034-0C-24N-010N-1N-0N: 8   PBI16C009-0C-0N-0N-010N-14N: 8
 (Other)                     :40   (Other)                     :39
                 WT     ASide  ABlock    WU.Total
 Limited-Ambient  :40   N:79   1:80   Min.   :0.4881
 Limited-Heat     :39   S:80   2:79   1st Qu.:1.5072
 Unlimited-Ambient:40                 Median :2.3507
 Unlimited-Heat   :40                 Mean   :2.7552
                                      3rd Qu.:3.9597
                                      Max.   :6.2256
                                      NAs    :2
```

```
anat.stoma.dat <- anat.stoma.dat[order(anat.stoma.dat$Cell.ID),]
summary(anat.stoma.dat)
```

```
     Filename       Stomata      StomatalArea  StomatalWidth   StomatalLength
 Length   :4189   1      : 159   Min.   : 428   Min.   :14.85   Min.   :31.85
 N.unique : 159   2      : 159   1st Qu.:1411   1st Qu.:30.59   1st Qu.:57.26
 N.blank  :   0   3      : 159   Median :1596   Median :33.41   Median :62.27
 Min.nchar:  42   4      : 159   Mean   :1619   Mean   :34.05   Mean   :61.84
 Max.nchar:  65   5      : 159   3rd Qu.:1793   3rd Qu.:36.72   3rd Qu.:66.46
                  6      : 159   Max.   :3140   Max.   :58.14   Max.   :99.83
                  (Other):3235   NAs    :15     NAs    :5       NAs    :9
 Pore length (µm) Pore width (µm)     PoreArea      Pore depth (µm)     Temperature
 Min.   : 17.29   Min.   : 8.647   Min.   : 234.9   Min.   : 4.324   Length   :4189
 1st Qu.: 31.10   1st Qu.:15.548   1st Qu.: 759.2   1st Qu.: 7.774   N.unique :   2
 Median : 33.82   Median :16.911   Median : 898.0   Median : 8.456   N.blank  :   0
 Mean   : 33.64   Mean   :16.819   Mean   : 897.9   Mean   : 8.409   Min.nchar:   4
 3rd Qu.: 36.10   3rd Qu.:18.049   3rd Qu.:1022.9   3rd Qu.: 9.025   Max.nchar:   7
 Max.   :104.84   Max.   :52.420   Max.   :2307.7   Max.   :26.210
                                   NAs    :9
     Watering          Rep       Line_Temperature_Watering_Rep Replicate    Line.ID
 Limited  :2115   Min.   :1.00   Length   :4189                1: 985    L06    : 488
 Unlimited:2074   1st Qu.:2.00   N.unique : 159                2:1059    L07    : 458
                  Median :3.00   N.blank  :   0                3:1085    L10    : 434
                  Mean   :2.53   Min.nchar:  16                4:1060    L08    : 424
                  3rd Qu.:4.00   Max.nchar:  22                          L04    : 417
                  Max.   :4.00                                           L09    : 412
                                                                         (Other):1556
  Rm_Temperature      Cell.ID           Pot.ID       Room      Lane       Position
 7,Heat   :2088   Length   :4189   Length   :4189   CER7:2088   A:736   11     : 340
 8,Ambient:2101   N.unique : 159   N.unique : 159   CER8:2101   B:600   5      : 338
                  N.blank  :   0   N.blank  :   0               C:751   10     : 325
                  Min.nchar:   7   Min.nchar:   6               D:713   3      : 323
                  Max.nchar:   7   Max.nchar:   6               E:660   2      : 318
                                                               F:729   9      : 317
                                                                       (Other):2228
    PCoord     Side      Location    Block    SubUnit       Cell          Treatment
 11     : 232   N:2087   Inner:2082   1:2060   1:2127   7      : 438   Length   :4189
 9      : 228   S:2102   Outer:2107   2:2129   2:2062   3      : 436   N.unique :  40
```

```
5       : 225                                          5       : 435   N.blank  :   0
3       : 224                                          2       : 429   Min.nchar:   3
10      : 218                                          6       : 426   Max.nchar:   3
7       : 214                                          8       : 420
(Other):2848                                           (Other):1605
                          Line          AWatering        HWatering
PBI16C009-0C-0N-0N-010N-14N: 488   rest     :2088   rest     :2101
PBI19N007-0N-3N            : 458   Limited  :1071   Limited  :1044
PBI17N002-0C-0N-0N-010N-16N: 434   Unlimited:1030   Unlimited:1044
PBI17N007-0C-0N-0N-010N-8N : 424
PBI17N013-0C-0N-0N-010N-8N : 417
PBI17N016-0C-0N-0N-010N-27N: 412
(Other)                    :1556
                        ALine                           HLine
rest                       :2088   rest                       :2101
PBI16C009-0C-0N-0N-010N-14N: 231   PBI16C009-0C-0N-0N-010N-14N: 257
PBI17N013-0C-0N-0N-010N-8N : 218   PBI19N007-0N-3N            : 240
PBI19N007-0N-3N            : 218   PBI17N002-0C-0N-0N-010N-16N: 220
PBI17N016-0C-0N-0N-010N-27N: 215   PBI17N007-0C-0N-0N-010N-8N : 211
PBI17N002-0C-0N-0N-010N-16N: 214   PBI17N013-0C-0N-0N-010N-8N : 199
(Other)                    :1005   (Other)                    : 961
                TW                         WT       ASide    ABlock      WU.Total
Ambient-Limited  :1071   Limited-Ambient  :1071   N:2044   1:2070   Min.   :0.4881
Ambient-Unlimited:1030   Limited-Heat     :1044   S:2145   2:2119   1st Qu.:1.5072
Heat-Limited     :1044   Unlimited-Ambient:1030                     Median :2.2327
Heat-Unlimited   :1044   Unlimited-Heat   :1044                     Mean   :2.7279
                                                                    3rd Qu.:3.9576
                                                                    Max.   :6.2256
                                                                    NAs    :49
```

Save data

```
save(anat.pot.dat, file="../data/anat.pot.dat.v2.rda")
save(anat.stoma.dat, file="../data/anat.stoma.dat.v2.rda")
save.image(saveFile)
```